

Langlands for SL2

Let us start with some useful results from representation theory:

Theorem 0.1: Clifford's Theorem

G be a group, $N \trianglelefteq G$ a normal subgroup, K a field, $\pi : G \rightarrow \mathrm{GL}_n(K)$ be an irreducible representation. Then the restriction $\pi|_N$ breaks up into a direct sum of irreducible representations of N of equal dimensions (i.e. it is semisimple and each representation is of the same dimension). These irreducible representations of N lie in one orbit for the action of G by conjugation on the equivalence classes of irreducible representations of N (i.e. they are isotypic components with the same multiplicity).

Proof :

Let W be an irreducible subrepresentation of $V|_H$. Since H is normal in G , for $g \in G$, H acts irreducibly on gW by $(h, gw) \mapsto hgh^{-1}hw$. The subspace $\sum_{g \in G} gW$ is a nonzero subrepresentation of V . Since V is irreducible, it is equal to V . Since a representation which is a sum of irreducible sub-representations is semisimple, $V|_H$ is semisimple. A similar argument follows for the 2nd assertion, completing the proof.

Let F be a locally compact nonarchimedean field with residue characteristic (ex. $F/\mathbb{Q}_p$), and $L = \overline{L}$ an algebraically closed field of characteristic $\neq p$. This paper will be about L -representations of $\mathrm{SL}_2(F)$. The reader can assume $L = \mathbb{C}$ to link this to the intuition complex representations like in [1]. The goal is to show how the Local Langlands correspondence goes from being a bijection to a surjection, with fibers of size 1, 2, or 4; these fibers are called *L-packets* or *L-indistinguishable's*.

Let $G = \mathrm{GL}_2(F)$, $G' = \mathrm{SL}_2(F)$, Z the center, G the torus, B the Borel subgroup, U the unipotent subgroup. If $X \subseteq G$ is a subset, $X' = X \cap G'$ and $x = (x_{ij})$ represents a matrix in G

Take $H = ZG'$. It is a closed normal compact group where G/H is isomorphic to $F^*/(F^*)^2$. It is normal as Z is obviously normal and $G' = \ker \det$ where $\mathrm{im} \det = F^*$, and $gZG'g^{-1} = ZG'$. It is closed since $H = \det^{-1}((F^*)^2)$. It is cocompact since:

$$\frac{F^*}{(F^*)^2} \cong \frac{\mathbb{Z}}{2\mathbb{Z}} \times \frac{\mathcal{O}_F^*}{(\mathcal{O}_F^*)^2}$$

and $\mathcal{O}_F^*$ is compact and so is the quotient by a closed subgroup. The $\mathbb{F}_2$ -dimension of G/H is that of $F^*/(F^*)^2$ which is:

$$\dim_{\mathbb{F}_2} F^*/(F^*)^2 = \begin{cases} 2 + e & \text{if } \mathrm{char}_F \neq 2, \\ \infty & \text{if } \mathrm{char}_F = 2, \end{cases} \quad \text{where } 2\mathcal{O}_F = \mathfrak{p}_F^e.$$

Hence if $\mathrm{char}_F \neq 2$, ZG' is open. Next, let $\overline{F}^{\mathrm{sep}}$ represent the separable algebraic closure, $\mathcal{W}_F$ and G_F the Weil group and galois group of $\overline{F}^{\mathrm{sep}}/F$ respectively. Let Ψ be a smooth additive L -character on F , trivial on $\mathcal{O}_F$ but not on $\mathfrak{p}_F^{-1}$ (i.e. a level 1 character, but not on $\mathfrak{p}_F^{-1}$ (i.e. A level 1 character)).

Definition 0.2: Whittaker Space

The space of smooth L -characters of U have the form

$$\theta(u) = \psi \circ \text{tr}(Y(u - 1)) = \Psi(Y_{2,1}, u_{1,2}) \quad u \in U$$